OTC Importance-Sampling Pricing Audit

Expected (Black-Scholes) vs Actual (IS Monte-Carlo)

Audits the importance-sampling MC pricer for illiquid alt-coin OTC derivatives. For each scenario we compare the analytic Black-Scholes price (expected) against drift-tilt IS and plain MC (actual), and verify that IS reduces variance — especially for deep-OTM strikes — and that pricing stays resilient through a simulated feed drop.

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Report Date:   (generic build)
Pricer:        scripts/_otc_fallback_pricer (bundled fallback)
MC Paths:      200,000   Seed: 7
Risk-Free:     3.0%
Scenarios:     5  (ATM / OTM / deep-OTM / digital / feed-drop)
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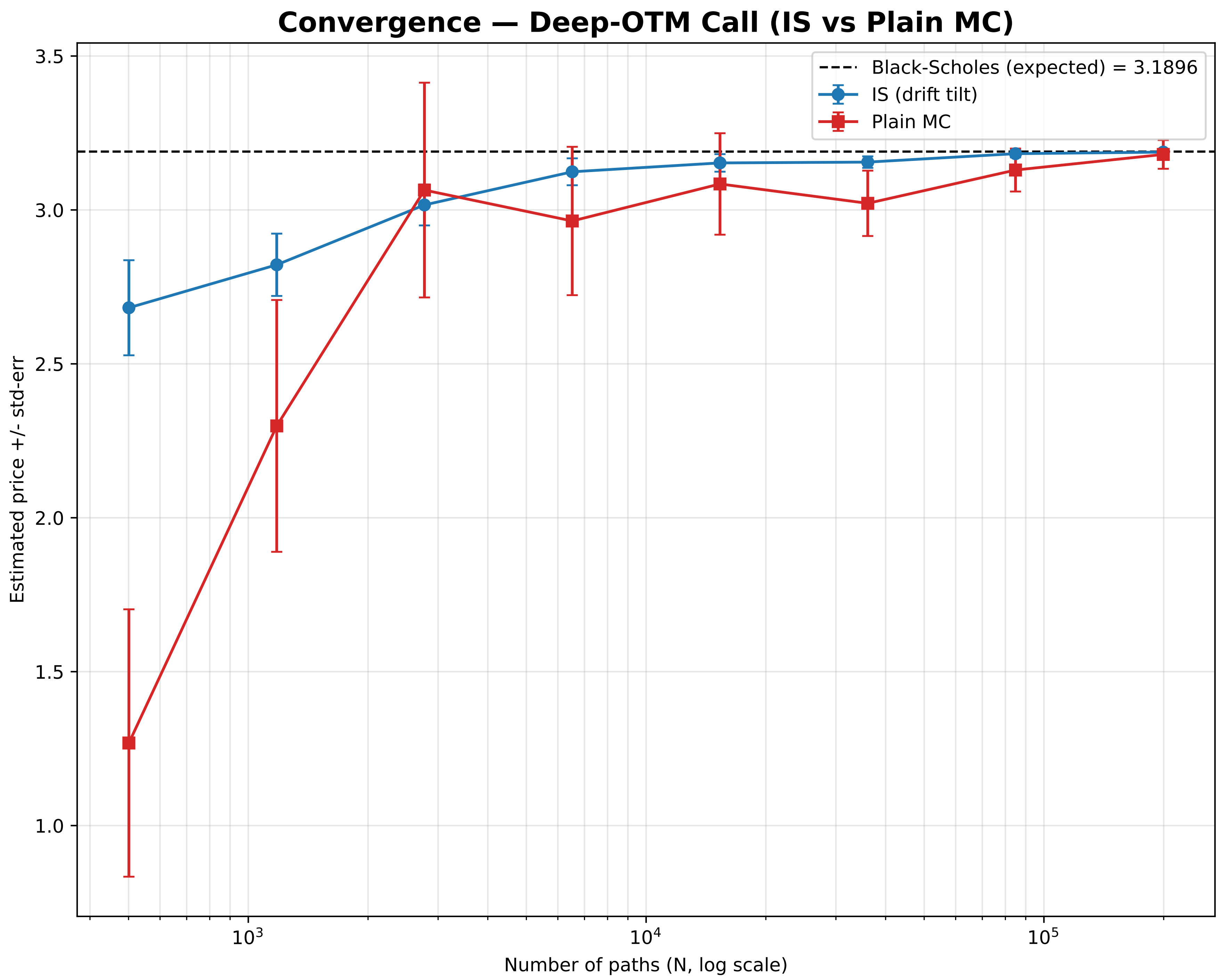
Expected vs Actual — Per Scenario

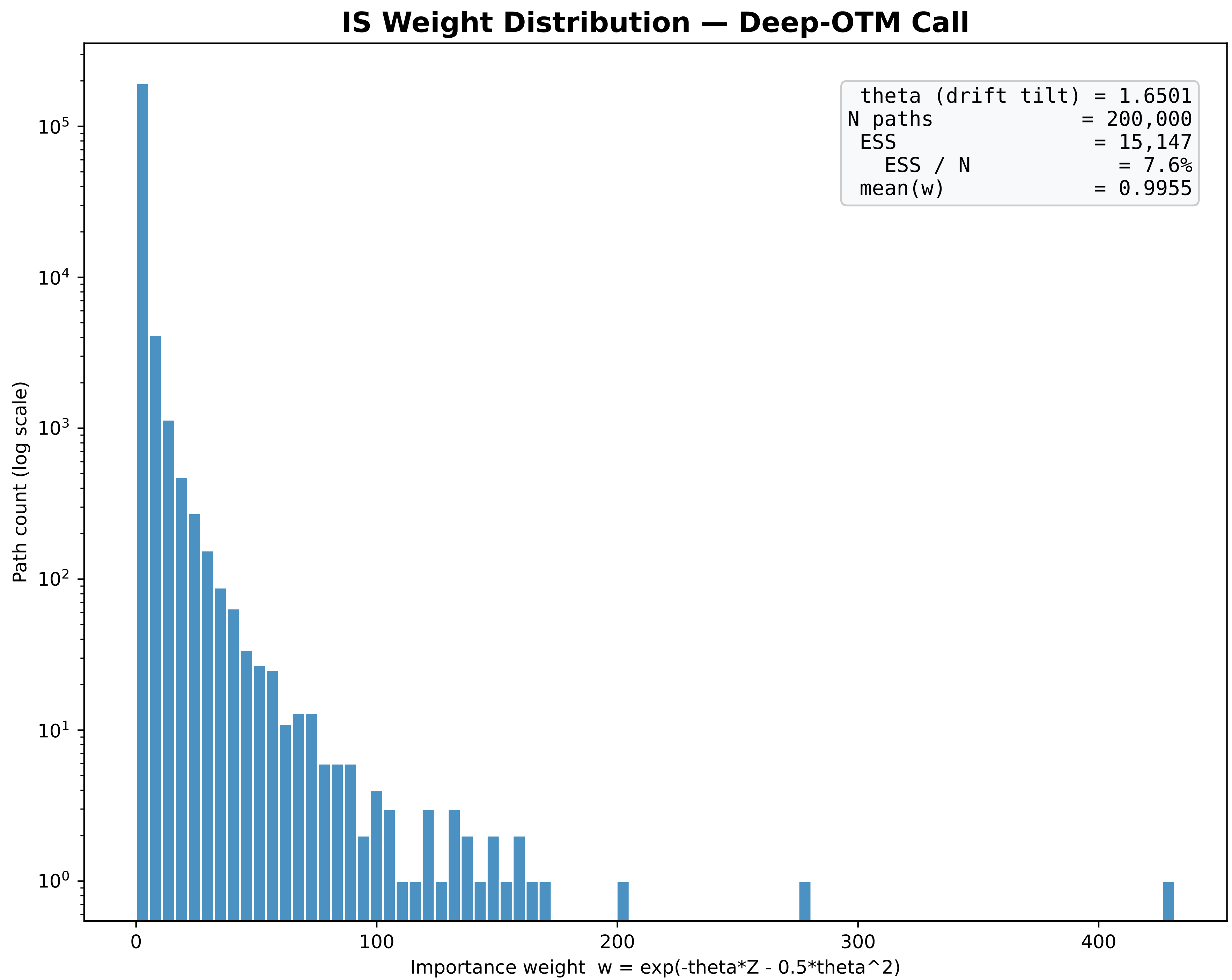
Expected = Black-Scholes analytic | Actual = drift-tilt IS MC

Scenario	Expected (BS)	Actual (IS)	Plain MC	IS std-err	Plain std-err	ESS	Var-red ratio
ATM call	22.8564	22.8472	22.8425	0.07738	0.10579	187,296	1.9x
OTM call	11.5477	11.5436	11.5387	0.03102	0.08136	97,174	6.9x
Deep-OTM call	3.1896	3.1880	3.1842	0.00792	0.04665	15,147	34.7x
Digital call	0.1628	0.1629	0.1632	0.00044	0.00082	78,071	3.5x
Feed-drop call	8.9821	8.9789	8.9778	0.02408	0.05837	107,092	5.9x

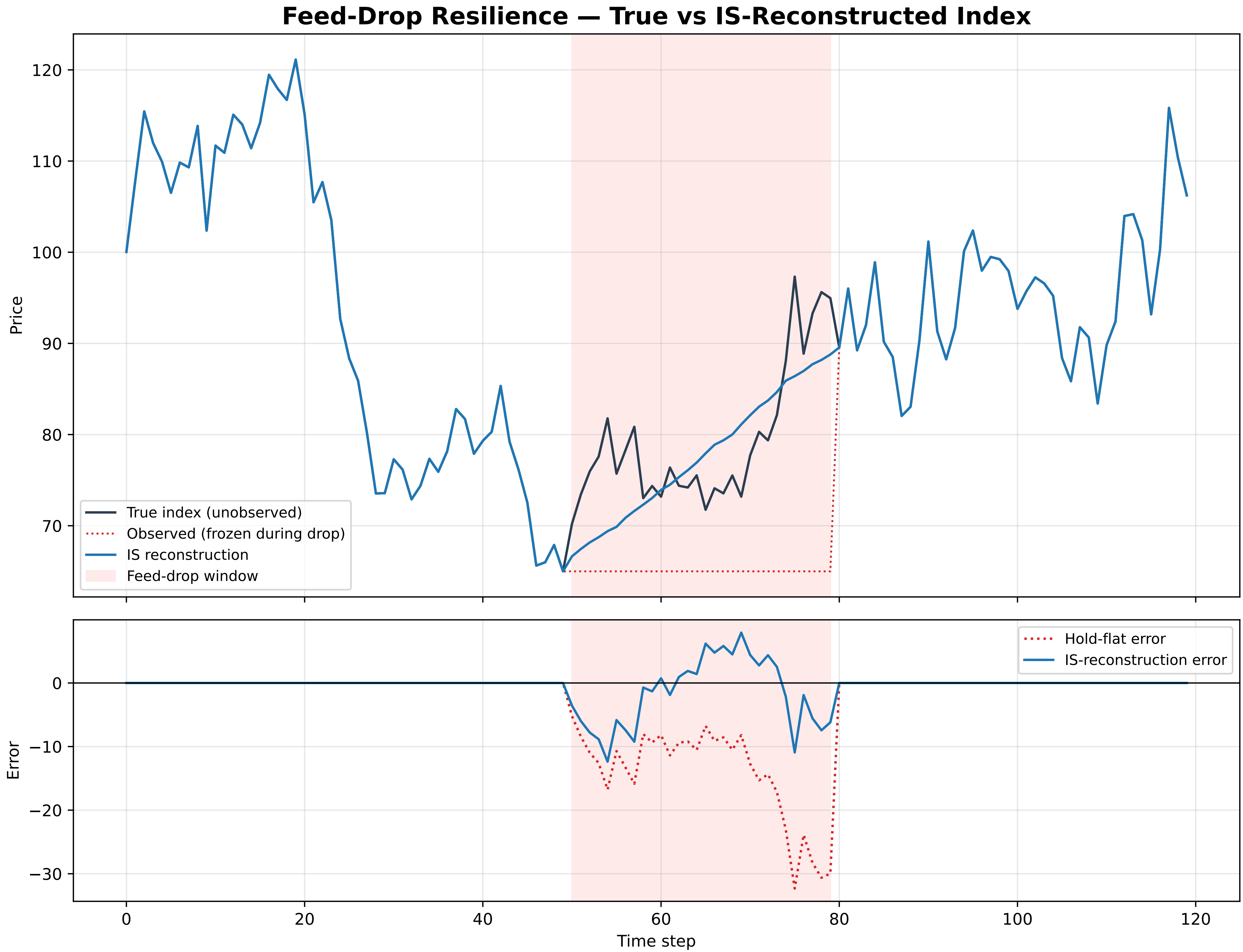
Var-reduction ratio = (plain-MC std-err / IS std-err)^2. Ratio > 1 means IS is more efficient; the gain grows for deeper OTM strikes.



Convergence — Deep-OTM Call (IS vs Plain MC)
Black-Scholes (expected) = 3.1896
IS (drift tilt)
Plain MC
Estimated price +/- std-err
Number of paths (N, log scale)
3.5
3.0
2.5
2.0
1.5
1.0
10^3
10^4
10^5



IS Weight Distribution — Deep-OTM Call
theta (drift tilt) = 1.6501
N paths = 200,000
ESS = 15,147
ESS / N = 7.6%
mean(w) = 0.9955
Path count (log scale)
Importance weight w = exp(-theta*Z - 0.5*theta^2)



Feed-Drop Resilience — True vs IS-Reconstructed Index
Price
120
110
100
90
80
70
True index (unobserved)
Observed (frozen during drop)
IS reconstruction
Feed-drop window
Error
0
−10
−20
−30
Hold-flat error
IS-reconstruction error
0
20
40
60
80
100
120
Time step